

120 Days of Quant Finance

Month 2: Derivatives & Time Series

Day 30: Options Greeks and Risk Measures (Part 2)

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February 2026

Introduction

In Part 1, option Greeks were introduced as sensitivity measures describing how an option's value responds to changes in market variables. In this section, we derive and interpret the primary Greeks under the Black–Scholes framework and examine their structural relationships.

These closed-form expressions form the foundation of modern option risk management and dynamic hedging.

Black–Scholes Option Pricing Model

Under the Black–Scholes model, the price of a European call option is given by:

$$C(S, t) = SN(d_1) - Ke^{-r(T-t)}N(d_2),$$

where

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}.$$

Here, S is the underlying price, K is the strike, T is the maturity, r is the risk-free interest rate, σ is volatility, and $N(\cdot)$ denotes the standard normal cumulative distribution function.

Delta under Black–Scholes

For a European call option, Delta is given by:

$$\Delta_{\text{call}} = \frac{\partial C}{\partial S} = N(d_1).$$

For a European put option:

$$\Delta_{\text{put}} = N(d_1) - 1.$$

Delta represents the probability-weighted exposure of the option to the underlying asset and plays a central role in hedging strategies.

Gamma under Black–Scholes

Gamma is identical for calls and puts and is given by:

$$\Gamma = \frac{\partial^2 C}{\partial S^2} = \frac{N'(d_1)}{S\sigma\sqrt{T-t}},$$

where $N'(d_1)$ is the standard normal probability density function.

Gamma measures the curvature of the option price and quantifies how rapidly Delta changes with the underlying price.

Theta under Black–Scholes

Theta measures sensitivity to the passage of time. For a European call option:

$$\Theta_{\text{call}} = -\frac{SN'(d_1)\sigma}{2\sqrt{T-t}} - rKe^{-r(T-t)}N(d_2).$$

Theta is typically negative for long options, reflecting time decay, and accelerates as maturity approaches.

Vega under Black–Scholes

Vega measures sensitivity to volatility and is identical for calls and puts:

$$\nu = \frac{\partial C}{\partial \sigma} = S\sqrt{T-t}N'(d_1).$$

Vega is highest for at-the-money options and increases with time to maturity, highlighting the importance of volatility in option valuation.

Rho under Black–Scholes

Rho measures sensitivity to interest rates. For a European call option:

$$\rho_{\text{call}} = \frac{\partial C}{\partial r} = K(T - t)e^{-r(T-t)}N(d_2).$$

Rho is generally small for short-dated equity options but becomes significant for long-dated contracts and interest-rate-sensitive derivatives.

Structural Relationships Between Greeks

Option Greeks are not independent. Their relationships emerge naturally from Taylor expansions of the option value:

$$dV \approx \Delta dS + \frac{1}{2}\Gamma dS^2 + \Theta dt + \nu d\sigma + \rho dr.$$

This decomposition provides a local approximation of option value changes and forms the basis of dynamic hedging and risk attribution.

Practical Interpretation

Delta captures directional exposure, Gamma captures curvature risk, Theta captures time decay, Vega captures uncertainty exposure, and Rho captures funding sensitivity. Together, these Greeks provide a complete first- and second-order description of option risk under the Black–Scholes model.

Conclusion

Closed-form expressions for Greeks under Black–Scholes allow precise quantification of option risk. These sensitivities are central to hedging, pricing, and portfolio construction and motivate more advanced models when market behavior deviates from Black–Scholes assumptions.